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EXPERT VIEW

Keep in touch: the soil–root hydraulic continuum and its role in drought resistance in crops

Pablo Affortit[1], Mutez Ali Ahmed[2], Alexandre Grondin[1], Silvain Delzon[3], Andrea Carminati[4], and Laurent Laplaze[1,*]

[1] DIADE, IRD, CIRAD, Université de Montpellier, Montpellier, France
[2] Root-Soil Interaction, School of Life Science, Technical University of Munich, Freising, Germany
[3] BIOGECO, INRAE, Université de Bordeaux, Pessac, France
[4] Department of Environmental Systems Science, ETH Zürich, Zürich, Switzerland

[*] Correspondence: laurent.laplaze@ird.fr

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Abstract

Drought is a major threat to food security worldwide. Recently, the root–soil interface has emerged as a major site of hydraulic resistance during water stress. Here, we review the impact of soil drying on whole-plant hydraulics and discuss mechanisms by which plants can adapt by modifying the properties of the rhizosphere either directly or through interactions with the soil microbiome.

Keywords: Arbuscular mycorrhizal fungi (AMF), extracellular polymeric substances (EPS), mucilage, rhizodeposition, rhizosheath, root architecture, root hairs.

Introduction

Water stress is the main factor affecting crop yield worldwide (Hammer *et al.*, 2021), and the impact of water stress on agriculture is expected to augment with increased frequency and intensity of drought spells predicted in most future climate scenarios (Potopova *et al.*, 2016; Fahad *et al.*, 2017). Understanding the factors that control water acquisition and use in plants is critical to adapt agriculture to future drier climates (Tardieu, 2022).

The hydraulic network at the whole-plant scale can be described using a demand and supply scheme (Fig. 1). Water loss occurs mostly in leaves by transpiration through stomata, small pores where gas exchange with the atmosphere occurs and whose opening is tightly regulated by the plant water status (Wang *et al.*, 2022). Transpiration is driven by the difference in vapour pressure between the stomatal cavity and the atmosphere, which ultimately creates a gradient of increasing bulk water potentials from the leaves to the roots. This gradient of water potential drives water from the soil within the plant only when the water potential of the root is more negative than that of the surrounding soil. In wet soils and under high transpiration demand, the largest drop in water potential is within the plant and, in these conditions, the main hydraulic resistance corresponds to the active regulation of transpiration by stomata. As the soil dries, an important resistance occurs in the soil surrounding the roots. In these conditions, plant traits that impact the water flow to the root surface are relevant for controlling leaf water potential and plant water status.

Here, we describe the impacts of the drop in soil matric potential on plant hydraulics in drying soil and discuss mechanisms by which plants cope with the reduction in soil hydraulic conductivity by modifying the properties of the rhizosphere.

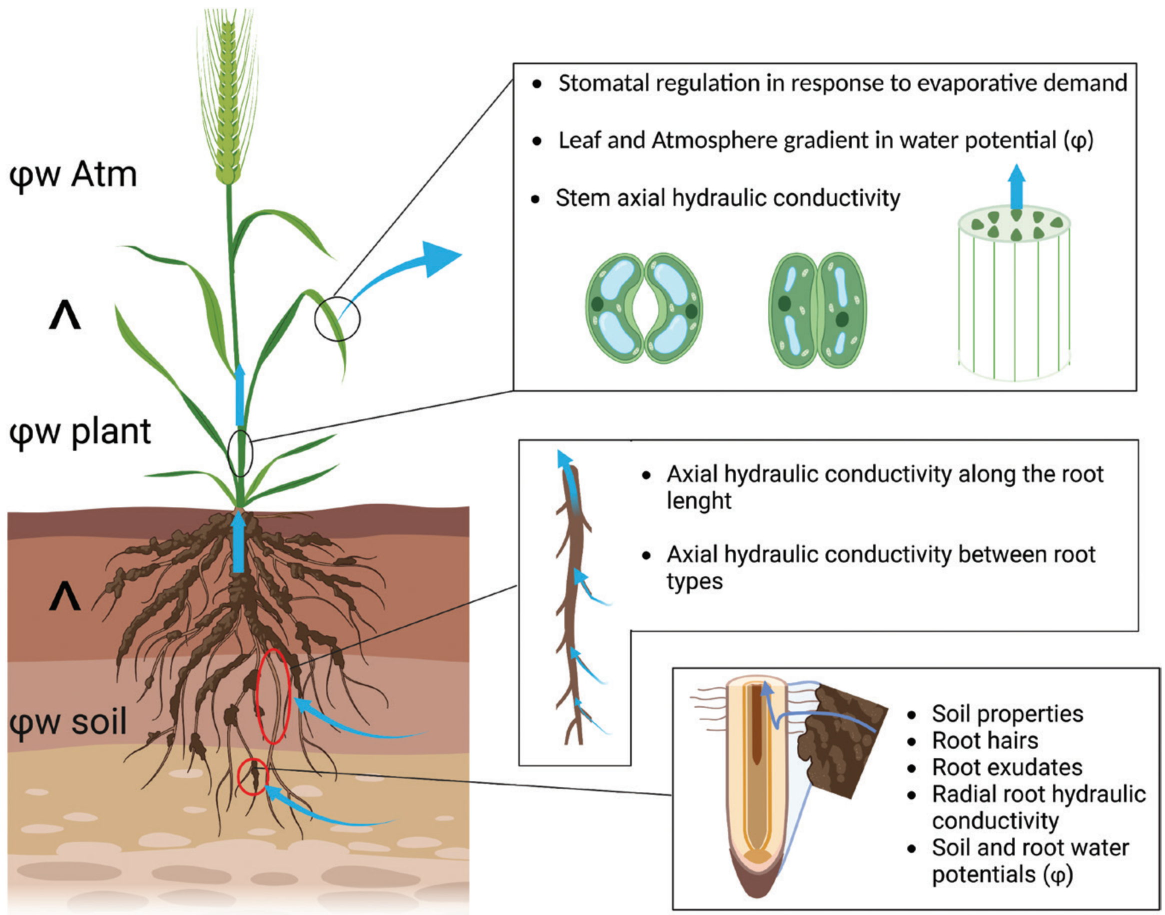


Fig. 1. The soil–plant–atmosphere continuum. Water movement is driven by differences in water potentials.

We further illustrate how a complex interplay between roots and the soil microbiome contributes to sustain the water flow across the soil–root interface under drought.

The soil–root hydraulic continuum and its impact on drought tolerance

Although the soil dries over weeks during a cropping season, millimetre-scale gradients of soil moisture develop around roots over minutes when the transpiration demand cannot be matched by the flow of water from the soil. These gradients are the result of the non-linearity of the soil hydraulic conductivity, which decreases by several orders of magnitude as the soil matric potential drops, and of the radial geometry of the flow towards roots. Gradients of water potential around the roots were predicted since the early work by Gardner (1960). The concept of matric flux potential allowed more accurate calculation of these gradients (e.g. van Lier *et al.*, 2008, 2013). Despite the improvements in modelling, these gradients remain challenging to measure. An experimental set-up developed by Passioura (1980) was used to estimate gradients in soil water potential toward the root from the deviation of the leaf water potential from its linear trajectory. Non-linearity in the relationship between transpiration and leaf water potential was interpreted as the consequence of a loss in the hydraulic conductivity (defined as the intrinsic ability of a material to conduct water, expressed in m s^{-1}) of the soil, of the root, or of the root–soil interface. We refer to the hydraulic conductance (defined as the relationship between conductivity and driving force, expressed in m s^{-1} MPa^{-1}—conductance is the inverse of resistance) of the root–soil continuum as the combination of the hydraulic conductance of the soil, the root, and their interface; that is, as it is a flow in series, it is the harmonic mean of the conductance of the three elements.

Plants continuously adapt to variable atmospheric and soil conditions by altering the hydraulic conductivity of key elements below and above ground. Although the occurrence of these alterations is well accepted, our quantitative understanding of this hydraulic acclimatization and its impact is incomplete. In wheat (*Triticum aestivum*), neither xylem collapse and cavitation nor a decrease in leaf conductance drive stomatal closure (Corso *et al.*, 2020). Instead, a decrease in soil–root hydraulic conductance was the main driver of stomatal closure during soil drying in tomato (Abdalla *et al.*, 2021) and in olive trees (Rodriguez-Dominguez and Brodribb, 2020; Box 1). Under drought, stomatal closure was controlled by a drop in soil water potential at the root–soil interface (Abdalla *et al.*, 2022). These results demonstrate that losses in root hydraulic conductance, which could be due to (i) a disconnection of the root from the soil during moderate water stress and/or (ii) collapse of root xylem conduits, are profound and sufficient to induce stomatal closure. In a recent review, Cai *et al.* (2022) investigated, across 11 crops and 10 contrasting soil textures, the effect of the interplay between soil and root hydraulic properties on water uptake during soil drying. The authors found that root water uptake was constrained within a wide range of soil water potential (−6 kPa to −1000 kPa), depending on both soil and root hydraulic properties (Cai *et al.*, 2022; Box 1). Furthermore, transpiration response to both soil drying and vapour pressure deficit was also found to be soil texture specific (Cai *et al.*, 2022; Koehler *et al.*, 2023; Box 1). The accumulation of salts at the root surface, which is likely to occur in soil drying conditions and at high transpiration rates, is an additional process hindering water flow across the root–soil interface. Salinity not only induced a more negative pre-dawn leaf water and reduced transpiration rate during soil drying, but also caused a reduction in root hydraulic conductance (Abdalla *et al.*, 2022).

Altogether, this shows that the root–soil interface is a major site of hydraulic resistance during drought. Accordingly, plants have developed ways of coping with the creation of gradients of soil water potential near the root during soil drying.

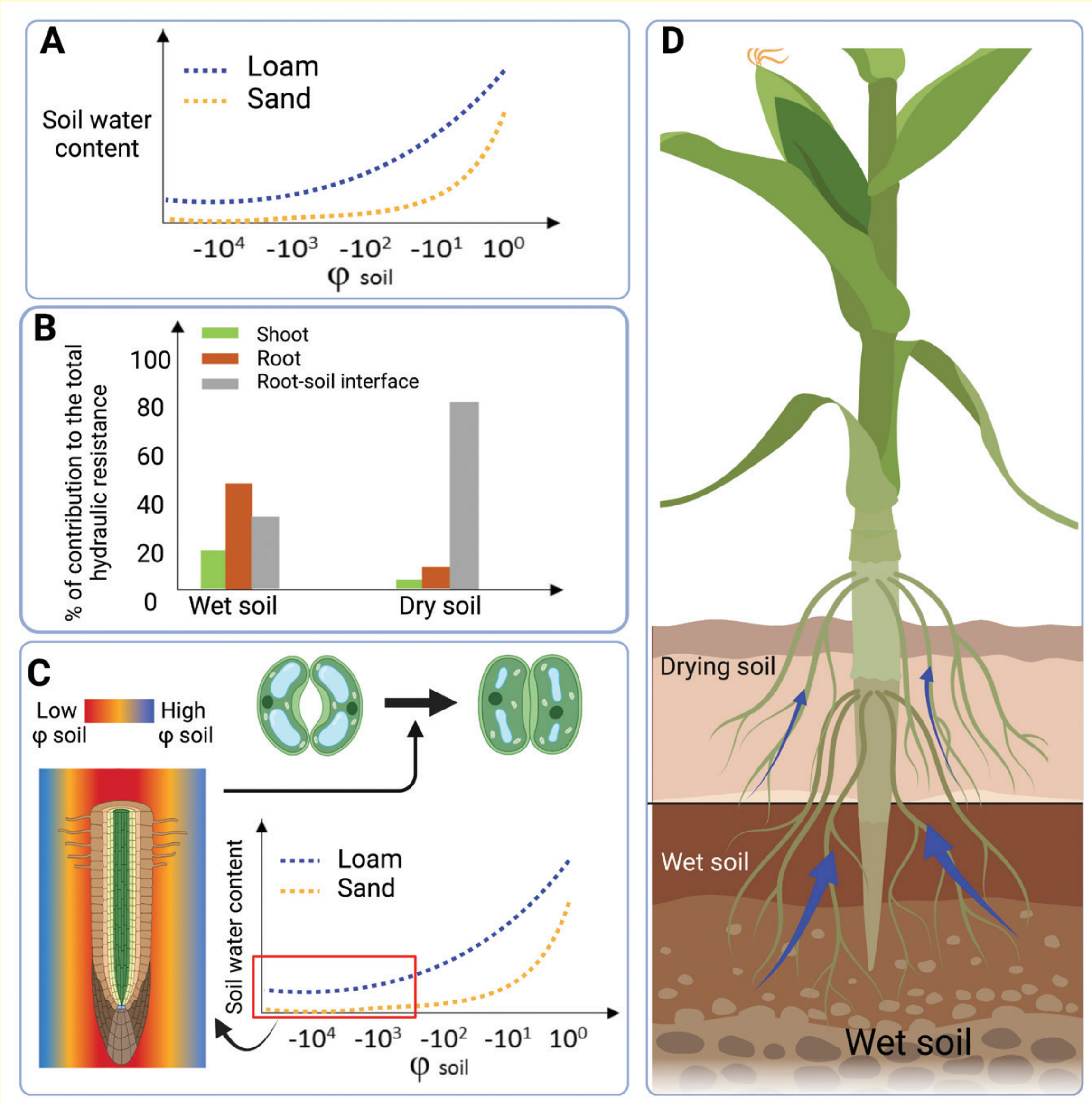
Root traits contribute to maintenance of the soil–root hydraulic continuum

Root architecture can impact water acquisition efficiency while root anatomy and physiology control the water transport capacity of the root (Lynch, 1995, 2022). Accordingly, root system architectures that would optimize water acquisition in different environments have been proposed based in large part on *in silico* simulations (e.g. Leitner *et al.*, 2014; Koch *et al.*, 2019; Lynch, 2022). Maintenance of the root–soil hydraulic continuum is important in root zones that actively acquire water. In maize, water uptake occurs mainly through seminal roots and their laterals at early developmental stages (Ahmed *et al.*, 2016b). As the root system develops, crown roots appear and become the main contributors of water uptake (Ahmed *et al.*, 2018). Similarly, crown roots have a higher contribution to water acquisition than primary roots in barley and wheat (Krassovsky, 1926; Sallans, 1942). Moreover, water uptake and water radial/axial hydraulic conductivity are not homogeneous along the root (Javaux, 2008). In maize, water acquisition occurs mainly in the apical parts in the seminal root (close to the tip; Zwieniecki, 2003). In addition, branching density as well as lateral root length impact the ability of the root to acquire water. For instance, a larger root surface area attenuates water potential gradients in the rhizosphere due to a lower water flux at the root–soil interface (Abdalla *et al.*, 2022; Cai *et al.*, 2022). Similarly, higher axial hydraulic conductivity in the apical segments combined with shorter and fewer laterals leads to a more uniform repartition of water potential along crown roots in maize (Ahmed *et al.*, 2018). Therefore, root water uptake is not homogeneous between and along root types, and the maintenance of the root–soil hydraulic continuum in drying soils appears crucial in crown root apical zones and highly branched regions.

Root traits have been associated with improved root–soil hydraulic continuity in conditions of water stress. Root hairs are tubular extensions of epidermal cells that increase the root surface area in contact with the soil (Ohlert, 1837; Haling *et al.*, 2013). The role of root hairs for nutrient acquisition is well documented, but their role in water uptake remains controversial (Cai and Ahmed, 2022). Barley plants with relatively long root hairs showed increased water uptake in drying soils (Carminati *et al.*, 2017; Marin *et al.*, 2021). On the other hand, no differences in water uptake were observed between the *rth3* root hairless maize mutant and its corresponding wild type (Cai *et al.*, 2021; Jorda *et al.* 2022). It was suggested that short root hairs (as in maize) have little contribution to root water uptake as opposed to relatively longer root hairs (as in barley; Cai and Ahmed, 2022). Barley roots also show denser root hairs as compared with maize which increase the soil–root contact and reduce the radial hydraulic resistance at the root–soil interface (Aslam *et al.*, 2022). Dense root hairs also influence soil density and porosity close to the root by aggregating soil particles (Hallett *et al.*, 2022). Vulnerability (shrinkage or distortion) of root hairs under water stress limits their potential role in water uptake in drying soils. Duddek *et al.* (2022) demonstrated that, in maize, root hair shrinkage (i.e. reduction of root hair length and volume) was initiated at relatively high soil matric potentials (between −10 kPa and −310 kPa; Box 2). Hence, recent evidence converges towards a predominant role for longer and denser root hairs in buffering the drop in matrix potential within the rhizosphere by maintaining the root–soil hydraulic continuum in drying soils (Carminati *et al.*, 2017; Marin *et al.*, 2021).

Beyond root hair formation, roots can influence the chemical and physical characteristics of their surrounding soil by releasing carbon compounds—a mechanism called rhizodeposition that includes root exudation and dead cell release from the root (Barber, 1995). Root exudates include high molecular weight compounds such as proteins and polysaccharides, and lower molecular weight compounds that are mainly composed of primary and secondary metabolites (amino acids, sugars, carboxylates, flavonoids, etc). Root exudates can account for 20–40% of the plant photosynthate depending upon species and varieties within species, and are highly controlled by environmental factors (de la Fuente Cantó *et al.*, 2020; Chen *et al.*, 2022). Quantitative and qualitative changes in rhizodeposition have been reported in response to drought (Preece and Peñuelas, 2016; Williams and Vries, 2020; Wang *et al.*, 2021). Drought often leads to a decrease in absolute rhizodeposition due to a decrease in photosynthesis and in carbon

Box 1. Recent developments in our understanding of the role of the rhizosphere properties in drought tolerance



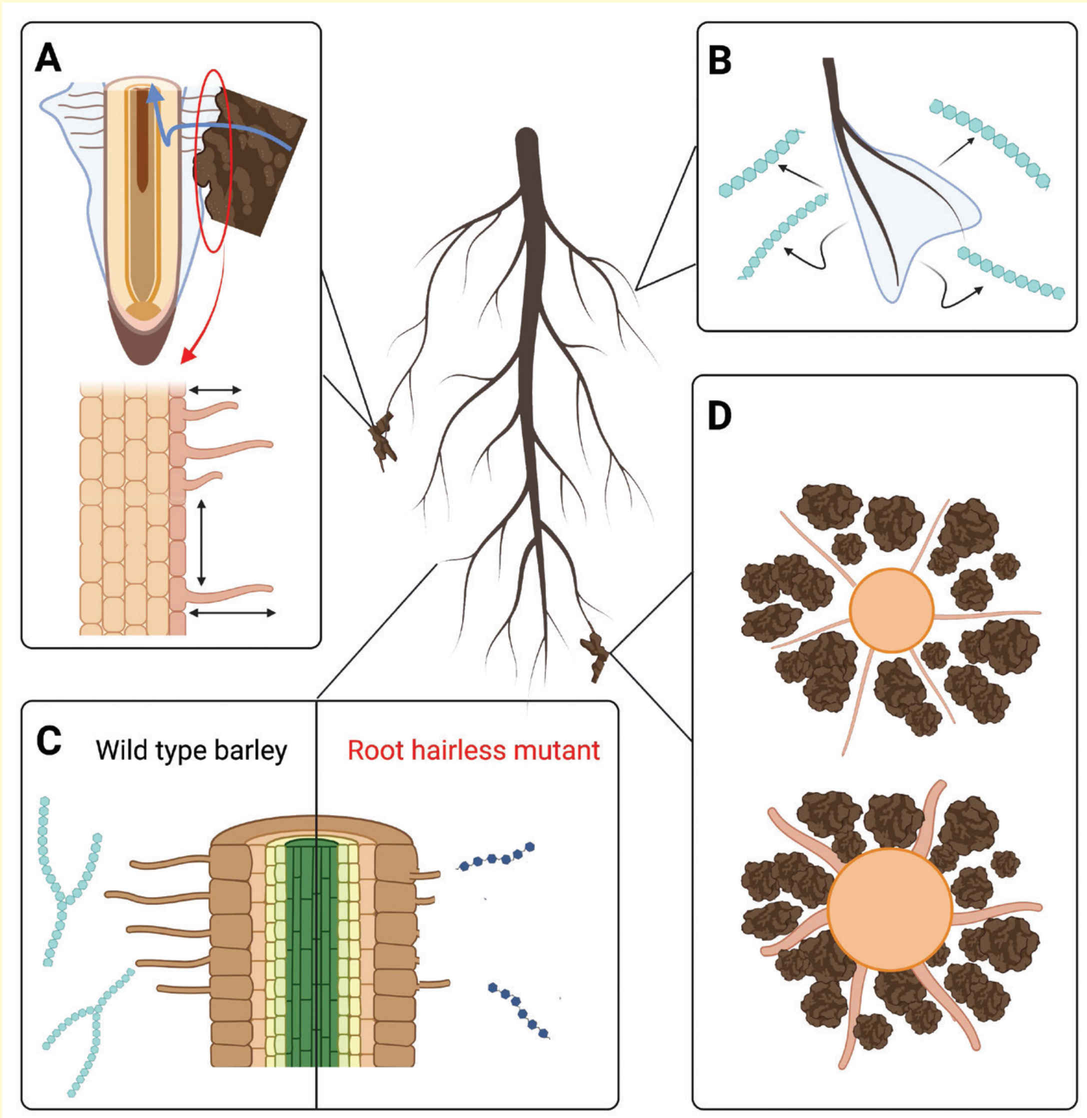
(A) Cai *et al.* (2022) showed that, during soil drying, root water uptake is constrained within a wide range of soil water potential (–6 kPa to –1000 kPa), depending on both soil and root hydraulic properties.

(B) Dominguez-Rodriguez and Brodribb (2020) demonstrated that upon soil drying, the hydraulic resistance of the root–soil interface increases to ~90% of the total soil–plant hydraulic resistance and thus represents the main constraint to transpiration. The resistance of each component was measured in olive trees by following the rehydration rate of plants/organs submitted to different drought stress intensities.

(C) Abdalla *et al.* (2021) illustrated that stomatal closure under drought is driven by an increase in soil–root hydraulic resistance.

(D) Müllers *et al.* (2023) estimated the vertical distribution of conductance and reported that, under drought, maize plants locally increase root conductivity in wetter soil layers, thereby compensating for a reduced root conductance in upper, drier layers.

Box 2. Key developments in our understanding of the role of the root traits that contribute to building up the rhizosphere



(A) Burak *et al.* (2021) highlighted that both root hairs length and density promote rhizosheath formation in barley and maize.

(B) Brax *et al.* (2020) studied the influence of the physico-chemical properties of root mucilage on the microstructural stability of sand.

(C) Galloway *et al.* (2022) showed altered properties and structures of polysaccharides exuded by the root in a root hairless mutant of barley. This highlighted the role of root hairs in rhizodeposition.

(D) Duddek *et al.* (2022) demonstrated that, in maize, root hair shrinkage during soil drying, was initiated at relatively high soil matric potentials (between –10 kPa and –310 kPa). This highlights the important role of root hairs in maintaining root–soil continuity.

availability for exudation. However, several studies suggest that there is an increase in carbon allocated to rhizodeposition relative to plant biomass in water stress conditions, highlighting the importance of this trait in response to drought (Williams and Vries, 2020; Wang *et al.*, 2021). Indeed, rhizodeposits can contribute to water stress tolerance directly by changing the physico-chemical properties of the rhizosphere or indirectly as a source of nutrients and signals for soil microbes.

Among rhizodeposits, mucilage comprise polysaccharides produced by active secretory cells of the root cap representing ~50% of the root exudates (Ropitaux *et al.*, 2020). Mucilage has a high water retention capability, thus enhancing water content around the root and possibly attenuating the drops in soil hydraulic conductivity during drying (Benard *et al.*, 2019). In addition, mucilage can bind soil particles together and thus stabilize soil aggregates (Morel *et al.*, 1991) and affect the physical stability of the rhizosphere (Brax *et al.*, 2020; Box 2). Mucilage deposition around the root tip might be particularly important to alleviate the drop in soil matric potential in this region of high water uptake (Carminati and Vetterlein, 2013; Carminati *et al.*, 2016). However, the impact of mucilage on water retention and hydraulic conductivity varies between species and is dependent on soil textures (Kroener *et al.*, 2018). For instance, modelling showed that coarse soils require a higher mucilage concentration to increase soil water content (Kroener *et al.*, 2018). The impact of mucilage also depends on the severity and frequency of drying episodes (Carminati and Vetterlein, 2013). Upon extreme soil drying, mucilage may create a protective layer preventing water loss from the root to the soil as it becomes water repellent when dried (Moradi *et al.*, 2011, 2012; Ahmed *et al.*, 2016a; Zickenrott *et al.*, 2016). Therefore, rhizodeposition of mucilage may play important roles in delaying soil drying of the rhizosphere and prevent root–soil loss of contact.

Altogether, these alterations of the soil by the plant can culminate in some species with the formation of a layer of soil adhering tightly to the root system, called the rhizosheath (McCully, 1999; George *et al.*, 2014; Ndour *et al.*, 2020; Aslam *et al.*, 2022). Rhizosheath formation is found in many plant families across the plant kingdom and depends mostly on root hair and rhizodeposition (Ndour *et al.*, 2020; Aslam *et al.*, 2022). In dry soils, the rhizosheath has higher water content than bulk soil, and may substantially contribute to water uptake (North and Nobel, 1997). Accordingly, a larger rhizosheath has been associated with improved resistance to water stress in chickpea, foxtail millet, and wheat (James *et al.*, 2016; Rabbi *et al.*, 2018; Liu *et al.*, 2019). Interestingly, rhizosheath formation is largely controlled by the plant genotype and could therefore be targeted by breeding programmes (George *et al.*, 2014; de la Fuente Cantó *et al.*, 2022).

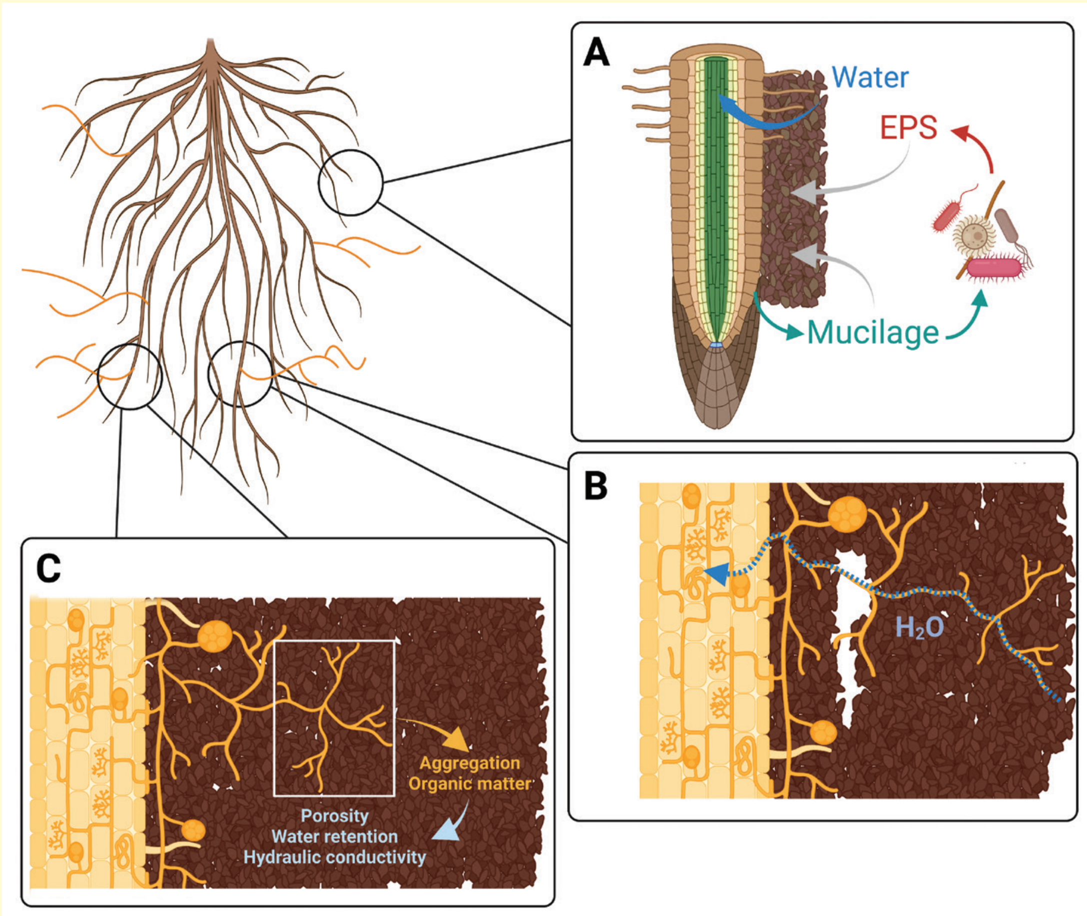
The interplay between root and soil microbiome contributes to shape the hydraulic properties of the rhizosphere

Roots interact with many organisms in the soil to shape the rhizosphere microbial communities (de la Fuente Cantó *et al.*, 2020). This in turn has a large impact on soil structure and hydraulic properties. In response to drought, photosynthesis is reduced and therefore limits carbon availability. In those conditions, there is a trade-off for carbon allocation between root development (root growth, branching, root hair development, etc.) and carbon invested in the soil compartment [rhizodeposition and symbiosis with soil microbes—chiefly arbuscular mycorrhizal fungi (AMF)], leading to different strategies to deal with water stress (Wang *et al.*, 2021).

Rhizodeposition is a major source of energy, carbon, and nitrogen for soil microbes and therefore has a very strong effect on the rhizosphere microbiota, both quantitatively (microbial biomass is positively correlated to the amount of rhizodeposits) and qualitatively (the structure of the microbial population influenced by the nature of the exudates; Preece and Peñuelas, 2016). Rhizodeposition varies both quantitatively and qualitatively along the root, between genotypes (species and varieties) and in response to environmental parameters such as water stress (Preece and Peñuelas, 2016; Williams and Vries, 2020; Wang *et al.*, 2021). Changes in rhizodeposition in response to water stress depend on the intensity of drought with a positive impact of mild stress, which decreases as stress intensity increases (Preece and Peñuelas, 2016). Crops seem to have more stable root exudation in response to drought stress than their wild relatives (Preece and Peñuelas, 2016).

The reduction in water availability within the soil as well as the changes in rhizodeposition brought about by water stress influence the rhizosphere microbiota (Preece and Peñuelas, 2016). Interestingly, the soil microbiota seems to be more resilient (i.e. its structure is less affected) to water stress than plants, and resilient microbiota that were subjected to previous water stress may improve plant resilience (Preece and Peñuelas, 2016). In some cases, changes in rhizodeposition and microbiota upon water stress are associated with improved plant drought tolerance (for reviews, see Preece and Peñuelas, 2016; Williams and Vries, 2020). Soil microbes shape the rhizosphere physico-chemical properties in many ways (de la Fuente Cantó *et al.*, 2020). Microbes produce extracellular polymeric substances (EPS) mainly composed of polysaccharides that are primarily involved in biofilm formation (Flemming and Wigender, 2010). Like mucilage, EPS bind soil particles together and change the hydraulic properties of the soil, improving water retention and maintaining the connectivity of the liquid phase around the root (Benard *et al.*, 2019; Nazari *et al.*, 2022; Box 3). This delays desiccation of the rhizosphere and helps to maintain a flow of water and nutrients during soil drying. In addition, microbial exopolysaccharides contribute to rhizosheath formation (Ndour *et al.*, 2020). It has been shown that bacteria from the root microbiota can modulate (either increase or decrease depending on the strain) suberin deposition in the root endodermis in the context of mineral nutrient homeostasis (Salas-Gonzales *et al.*, 2021). This would also influence root hydraulics as it creates a barrier for apoplastic water flow. The microbiome can also change root architecture, for instance by inducing root branching (Gonin *et al.*, 2023). This would also change root and rhizosphere

Box 3. Recent developments in our understanding of the role of the microbiome in shaping the root–soil hydraulic continuum



(A) Nazari *et al.* (2022) reported a meta-analysis of >80 studies showing the role of mucilages in shaping the rhizosphere microbial habitat. Mucilages are degraded by microbes that produce extracellular polymeric substances (EPS) with similar properties. Altogether, plant mucilage and bacterial EPS form a mucigel that binds soil particles together and changes the hydraulic properties of the soil.

(B) Kakouridis *et al.* (2022) showed that AMF hyphae can directly transport water from the soil to the root and that this transport can occur across air gaps in the soil. Part of the transport occurs in hyphae through an extracytoplasmic pathway.

(C) Pauwels *et al.* (2023) demonstrated that colonization by AMF hyphae changes water retention and hydraulic conductivity in root-free soil. However, the response was opposite in soils with very contrasted texture (loam and sand).

hydraulics as a larger root surface area attenuates water potential gradients in the rhizosphere. However, these effects of the root microbiome have not been studied in the context of water stress.

AMF have been shown to be the most important component of the root microbiome for drought tolerance in trees (Allsup *et al.*, 2023). The arbuscular mycorrhizal symbiosis is a mutualistic association between most angiosperms and AMF—from the Glomeromycota phylum (Redecker *et al.*, 2013). The fungus develops specific structures, called arbuscules, in root cortical cells to exchange resources with the plant and a network of hyphae that forage the soil for water and nutrients (Smith and Smith, 2011). Association with AMF has been known for a long time to improve plant resistance to water stress (Augé, 2001). Part of this effect is indirect via improved plant mineral nutrition and osmoregulation, reduced drought-induced oxidative stress, and increased root hydraulic conductivity (Mbodj *et al.*, 2018; Quiroga *et al.*, 2019; Zulfiqar and Ashraf, 2021). Moreover, the hyphal network binds soil particles together, stabilizes aggregates, impacts soil porosity (Miller and Jastrow, 2000; Hallett *et al.*, 2009), and links roots to the surrounding soil, limiting hydraulic continuity loss and the formation of air gaps during drying (Augé, 2001). However, the effect of AMF on soil water retention and hydraulic conductivity is very dependent on the soil texture, with opposite effects reported for loam and sand (Pauwels *et al.*, 2023; Box 3).

Recently, direct contribution of AMF to water transfer from the soil to plant roots via fungal hyphae was observed (Kakouridis *et al.*, 2022; Box 3). AMF hyphae can transport water across air gaps in the soil (Kakouridis *et al.*, 2022) and extract water from small pores that are inaccessible to roots, two properties that can be advantageous for the water scavenging in dry soils. Under edaphic stress, AMF maintained the hydraulic continuity between root and soil in drying soils, thereby reducing the drop in soil matric potential near the root surface and sustaining root water uptake in dry soils (Abdalla and Ahmed, 2021). Altogether, this indicates that AMF act as an extension of the root and increase access to soil resources including water.

Conclusion and outlook

The root–soil interface is an important hydraulic resistance upon soil drying, affecting whole-plant hydraulics and response to drought stress. Accordingly, plants have evolved different mechanisms to shape the physico-chemical properties of the rhizosphere to maintain the soil–root hydraulic continuum and water acquisition. This includes root hair development, rhizodeposition, and symbioses with AMF. Interestingly, all these traits are under plant genetic control and could therefore represent new breeding targets to improve plant water use and drought tolerance. Indeed, as recently demonstrated for maize, most of the past selection for drought tolerance was based on traits such as phenology and architecture, but did not exploit mechanistic traits that contribute to environmental adaptation such as rhizosphere traits (Welcker *et al.*, 2022). More work is needed to better characterize the genetic and physiological mechanisms controlling rhizosphere hydraulics and to evaluate how these impact drought tolerance in a breeding context. As roots influence the hydraulic properties of the rhizosphere, root positioning in the soil column (i.e. root system architecture) defines which regions of the soil exhibit root-induced alterations to their hydraulic properties and therefore the flow of water through the soil. Root system architecture and rhizospheric traits therefore need to be integrated to optimize water uptake by the plant.

Hydraulic properties of the soil–root interface can impact plant transpiration and eventually plant water use efficiency. In the actual and future context of water scarcity, agriculture needs to be more water efficient. Developing more efficient crops requires a holistic understanding of water use efficiency that involves a complex regulation of parameters related to soil properties, root traits, transpiration, and photosynthesis (Vadez *et al.*, 2023). To have an impact, these properties have to be regarded in the context of specific varieties used in different agrosystems prone to different stress patterns. Models combining soil–plant–atmosphere interactions at the plot level are needed to identify strategies better suited to different types of stress under variable agronomic contexts.

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Author contributions

All authors contributed to the first version of the text and revised the manuscript; PA, AG, and LL created the figures.

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Conflict of interest

The authors have no conflicts to declare.

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